deviceQualityDataManagement HTTP/TLS/JSON

Interface Design Description

Service ID: *"deviceQualityDataManagement"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `deviceQualityDataManagement` service of the DeviceQoSEvaluator system.

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1 Overview

This document describes the `deviceQualityDataManagement` service interface of the DeviceQoSEvaluator system, which provides operator access to the stored device statistics and rebuilding of the device-system measurement map.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `deviceQualityDataManagement` service interface

This document provides the Interface Design Description IDD to the *deviceQualityDataManagement – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `deviceQualityDataManagement` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https`, policy `CERT_AUTH`, access address `device-qos-evaluator`, port 8472 and base path `/deviceqosevaluator/mgmt`. The `operations` property of the interface lists the operations below with their method and path relative to the base path (path parameters are placeholders such as `{id}` that the consumer fills in):

Operation	Method	Path
`query`	POST	`/query`
`reload`	POST	`/reload`

2 Service Operations

The interfaces of the `deviceQualityDataManagement` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `deviceQualityDataManagement` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json` unless stated otherwise. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management, authorization or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

2.1 POST /deviceqosevaluator/mgmt/query

Operation: query
Input: QoSDeviceStatQueryRequest
Output: QoSDeviceStatQueryResponse

Returns a page of aggregated statistic records.

A successful call returns `200 OK` with the output in the body.

```
POST /deviceqosevaluator/mgmt/query HTTP/1.1
Host: device-qos-evaluator:8472
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "metricGroup": "RTT",
  "from": "2026-10-22T12:00:00Z",
  "to": "2026-10-22T12:00:00Z",
  "aggregation": [
    "string"
  ],
  "systemNames": [
    "TemperatureSensor"
  ]
}
```

Listing 1: An query invocation.

```
{
  "records": [
    {
      "metricGroup": "RTT",
      "id": 42,
      "timestamp": "string",
      "uuid": "string",
      "systems": [
        "string"
      ]
    }
```

```
  ],
  "count": 1
}
```

Listing 2: An query response.

2.2 POST /deviceqosevaluator/mgmt/reload

Operation: reload
Output: String

Rebuilds the device-system measurement map from the ServiceRegistry.
A successful call returns `200 OK` with the output in the body.

```
POST /deviceqosevaluator/mgmt/reload HTTP/1.1
Host: device-qos-evaluator:8472
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 3: An reload invocation.

```
"string"
```

Listing 4: An reload response.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.6, which are used to represent things like hashes, identifiers and texts.

3.1 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.2 struct Page

Paging and sorting; all fields optional.

Object Field	Value Type	Description
"page"	Number	Zero-based page number.
"size"	Number	Number of records per page (at most `max.page.size`, default 1000).
"direction"	String	Sort direction, `ASC` or `DESC`.
"sortField"	String	Field to sort by.

3.3 struct QoSDeviceStatQueryRequest

Paged filter for device statistics.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"metricGroup"	String	Metric group: `RTT`, `CPU_TOTAL_LOAD`, `MEMORY_USED`, `NETWORK_EGRESS_LOAD` or `NETWORK_INGRESS_LOAD`.
"from"	DateTime	Lower bound of the entry time.
"to"	DateTime	Upper bound of the entry time.
"aggregation"	Array<String>	Statistics to return (minimum, maximum, mean, median, current).
"systemNames"	Array<String>	Systems whose devices are returned.

3.4 struct QoSDeviceStatQueryResponse

Device statistic records.

Object Field	Value Type	Description
"records"	Array<QoSDeviceStatRecord>	Statistic records of the page.
"count"	Number	Total number of matching records (all pages).

3.5 struct QoSDeviceStatRecord

Aggregated measurements of one device in one period.

Object Field	Value Type	Description
"metricGroup"	String	Metric group: `RTT`, `CPU_TOTAL_LOAD`, `MEMORY_USED`, `NETWORK_EGRESS_LOAD` or `NETWORK_INGRESS_LOAD`.
"id"	Number	Identifier of the record.
"timestamp"	String	End of the aggregation period.
"uuid"	String	Identifier of the measured device.
"minimum"	Object	Minimum value in the period.
"maximum"	Object	Maximum value in the period.
"mean"	Object	Mean value in the period.
"median"	Object	Median value in the period.
"current"	Object	Last measured value.
"systems"	Array<String>	Systems hosted by the device.

3.6 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “deviceQualityDataManagement – Service Description (DeviceQoSEvaluator),” Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-deviceQualityDataManagement-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	